A Comparison of Temporal Data-Splitting Schemes for PPO-Based Portfolio Allocation

Aungruk Vanichanai
Faculty of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
67070297@kmitl.ac.th

Napat Wanitwatthakorn
Faculty of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
67070226@kmitl.ac.th

Nont Kanungsukkasem
Faculty of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
nont.ka@kmitl.ac.th

Anuntapat Anuntachai
Faculty of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
anuntapat.an@kmitl.ac.th

***Abstract*—This study investigates and compares three temporal data-splitting architectures (Single Holdout, Expanding Window, and Sliding Window) for multi-asset portfolio allocation using Proximal Policy Optimization (PPO) across SPY, QQQ, DIA, TLT, GLD, and cash under a two-tier risk control overlay (Volatility Targeting and Soft Drawdown Dampener). The empirical evaluation is structured into three phases: backtesting on a common out-of-sample period (Common OOS: 2024–2026), a 5-fold walk-forward analysis (2016–2026), and a 5-day live forward test executed via the Webull OpenAPI Sandbox. Initial backtest experiments report Common OOS Sharpe Ratios of 1.99, 2.01, and 2.44 for the Single Holdout, Expanding Window, and Sliding Window schemes, respectively, with the Sliding Window demonstrating superior resilience during the 2022–2023 macroeconomic regime shift. Conversely, live paper trading reveals that while all models achieved a net account appreciation of approximately 0.85 percent, their allocations degenerated into near-equal weights (approximately 20 percent per risky asset) with zero subsequent rebalancing after the initial trading day. A forensic system audit uncovers two critical implementation flaws: (1) feeding Z-score normalized Federal Funds Effective Rates (FEDFUNDS) directly into the financial accounting module for cash yield computation, and (2) an evaluation callback synchronization mismatch that saved model checkpoints at 4,000 timesteps prior to the first policy gradient update (zero updates), rendering the deployed agents functionally equivalent to randomly initialized policies. These findings underscore that rigorous verification of financial accounting mechanics and training loop synchronization is strictly prerequisite to interpreting quantitative performance metrics.**

Keywords—deep reinforcement learning, proximal policy optimization, portfolio allocation, temporal data splitting, backtesting, forward testing, quantitative finance

I. Introduction

A. Background and Motivation

The application of Deep Reinforcement Learning (DRL) [1] to dynamic portfolio allocation has garnered substantial attention due to the capacity of autonomous agents to learn adaptive trading policies across evolving market conditions. Nevertheless, the deployment of DRL in financial time series is fundamentally challenged by non-stationarity and data snooping bias [2]. Financial markets exhibit time-varying statistical properties driven by macroeconomic regime shifts; consequently, policies parameterized over a single historical partition frequently overfit to transient market dynamics and fail to generalize out-of-sample.

In evaluating policy gradient algorithms such as Proximal Policy Optimization (PPO) [3], the temporal data-splitting methodology serves as a critical architectural determinant governing both policy stability and knowledge retention. While existing literature predominantly relies on a single static train-test partition (Single Holdout)—which is highly susceptible to sample selection bias—walk-forward validation schemes, implemented via either an Expanding Window (anchored training set) or a Sliding Window (rolling fixed-length window), provide a more rigorous framework for assessing cross-regime robustness across diverse economic cycles [4].

B. Research Questions

This study addresses three primary research questions (RQs):

- **RQ1:** When evaluated over an identical out-of-sample horizon (Common OOS: 2024–2026), how do the risk-adjusted return profiles differ across Single Holdout, Expanding Window, and Sliding Window schemes?
- **RQ2:** How does the cross-regime robustness of Expanding Window and Sliding Window schemes vary across five sequential walk-forward folds spanning 2016 to 2026?
- **RQ3:** Does the live asset allocation behavior during forward testing align with historical backtest trajectories, and is the empirical evidence sufficient to validate the agency of the trained policies?

C. Research Contributions

1) *Controlled Architectural Comparison:* We benchmark all three temporal partitioning schemes under an identical experimental pipeline, incorporating standardized feature engineering, a 15-basis-point (bps) transaction cost model, and a unified two-tier volatility targeting overlay.
2) *Live Broker Sandbox Integration:* We engineer an automated live execution engine interfaced with the Webull OpenAPI Sandbox, complete with a persistent cross-day risk state synchronization module (`state_manager.py`).

3) *Forensic System Audit:* We document critical, often-overlooked software engineering pitfalls in financial DRL—specifically, a cash-yield feature normalization leakage bug and a premature checkpoint selection trap in multiprocessing Stable-Baselines3 [5] environments—offering actionable diagnostic protocols for the quantitative machine learning community.

II. Problem Formulation

The multi-asset portfolio allocation problem is formulated as a discrete-time Markov Decision Process (MDP).

A. State Space ($\mathcal{S}$)

At each discrete time step t, the environment state is represented by a 93-dimensional continuous vector $s_t \in \mathbb{R}^{93}$ comprising three distinct informational components:

$$s_t = \left[z_{t,1}, z_{t,2}, \dots, z_{t,K}, w_t^{\text{obs}}, s_t^{\text{risk}}, \frac{\sigma_t^{\text{realized}}}{\sigma^{\text{target,daily}}} \right] \in \mathbb{R}^{93} \quad (1)$$

where $K = 5$ denotes the number of risky Exchange-Traded Funds (ETFs): SPY, QQQ, DIA, TLT, and GLD; $z_{t,k} \in \mathbb{R}^{17}$ represents the standardized feature vector for asset k ($5 \times 17 = 85$ dimensions); $w_t^{\text{obs}} \in \mathbb{R}^6$ denotes the realized pre-open portfolio weights across all assets including the cash allocation $w_{t,0}$; $s_t^{\text{risk}} \in \mathbb{R}$ is the most recent risk exposure scalar; and $\sigma_t^{\text{realized}} / \sigma^{\text{target,daily}} \in \mathbb{R}$ is the ratio of realized portfolio volatility to the daily target volatility.

B. Train-Only Z-Score Normalization

To strictly prevent lookahead bias and data leakage, the sample mean and standard deviation for feature standardization are computed exclusively over the training partition ($\mathcal{T}_{\text{train}}$) of each respective fold:

$$z_{t,j} = \text{clip}\left(\frac{x_{t,j} - \mu_j^{\text{train}}}{\sigma_j^{\text{train}} + \epsilon_{\text{norm}}}, -\sigma_{\text{clip}}, \sigma_{\text{clip}} \right) \quad (2)$$

$$\mu_j^{\text{train}} = \frac{1}{|\mathcal{T}_{\text{train}}|} \sum_{t \in \mathcal{T}_{\text{train}}} x_{t,j} \quad (3)$$

$$\sigma_j^{\text{train}} = \sqrt{\frac{1}{|\mathcal{T}_{\text{train}}| - 1} \sum_{t \in \mathcal{T}_{\text{train}}} \left(x_{t,j} - \mu_j^{\text{train}} \right)^2} \quad (4)$$

where $x_{t,j}$ is the raw value of feature $j \in \{1, \dots, 17\}$ at time t; μ_j^{train} is the arithmetic mean over the training horizon; σ_j^{train} is the sample standard deviation over the training horizon; $\epsilon_{\text{norm}} = 10^{-8}$ is a numerical stability constant preventing division by zero; and $\sigma_{\text{clip}} = 5.0$ is the outlier clipping threshold.

C. Softmax Action Mapping

The actor network outputs a continuous unnormalized action vector $a_t \in [-3.0, 3.0]^K$, which is projected onto the standard simplex representing relative risky asset allocations via the Softmax function:

$$p_{t,i} = \frac{\exp(a_{t,i})}{\sum_{k=1}^{K} \exp(a_{t,k})}, \quad \forall i \in \{1, \dots, K\} \quad (5)$$

subject to the long-only simplex constraints $\sum_{i=1}^{K} p_{t,i} = 1.0$ and $p_{t,i} \geq 0$.

D. Two-Tier Volatility Targeting and Soft Drawdown Dampener

The risk engine estimates ex-ante portfolio variance using an Exponentially Weighted Moving Average (EWMA) model with decay factor $\lambda_{\text{ewma}} = 0.94$:

$$\sigma_t^2 = \lambda_{\text{ewma}} \sigma_{t-1}^2 + (1 - \lambda_{\text{ewma}}) \left(r_{t-1}^{\text{port}} \right)^2 \quad (6)$$

$$\sigma_t^{\text{realized}} = \sqrt{\max(\sigma_t^2, 10^{-8})} \quad (7)$$

where σ_t^2 is the estimated daily portfolio variance and r_{t-1}^{port} is the realized net portfolio return over the preceding period. The raw risk exposure scalar is computed by coupling target volatility scaling with a non-linear Soft Drawdown Dampener $\psi(DD_t)$:

$$s_t^{\text{raw}} = \left(\frac{\sigma^{\text{target,daily}}}{\sigma_t^{\text{realized}}} \right) \times \psi(DD_t) \quad (8)$$

$$\psi(DD_t) = 1.0 - \min\left(0.50,\ 2.0 \max(0,\ DD_t - 0.15)\right) \quad (9)$$

$$DD_t = \max\left(0,\ 1.0 - \frac{V_t}{V_t^{\text{peak}}} \right) \quad (10)$$

where $\sigma^{\text{target,daily}} = 0.10/\sqrt{252} \approx 0.0063$ corresponds to a 10% annualized volatility target; V_t and V_t^{peak} denote the current net liquidation value and the historical high-water mark of the portfolio, respectively; and $\psi(DD_t)$ linearly penalizes portfolio exposure when the current drawdown exceeds 15%, up to a maximum reduction of 50%.

To prevent excessive leverage and suppress microstructure-induced rebalancing noise, s_t^{raw} is bounded within [0.20,1.00] and filtered through a hysteresis band of width $\delta_{\text{hyst}} = 0.05$. With $c_t = \text{clip}(s_t^{\text{raw}}, 0.20, 1.00)$,

$$s_t = \begin{cases} s_{t-1}, & \text{if } |c_t - s_{t-1}| < 0.05 \\ c_t, & \text{otherwise} \end{cases} \quad (11)$$

The final target weights for the risky assets ($w_{t,i}$) and the residual cash buffer ($w_{t,0}$) are determined by:

$$w_{t,i} = s_t \cdot p_{t,i}, \quad i = 1, \dots, K \quad (12)$$

$$w_{t,0} = \max\left(0,\ 1.0 - \sum_{i=1}^{K} w_{t,i} \right) \quad (13)$$

E. Turnover Control and Financial Accounting

Single-way risky asset turnover is evaluated against the drifted pre-rebalancing weights $w_{t-1,i}^{\text{drifted}}$:

$$\tau_t^{\text{cand}} = \sum_{i=1}^{K} \left| w_{t,i} - w_{t-1,i}^{\text{drifted}} \right| \quad (14)$$

$$\tau_t = \begin{cases} 0.0, & \text{if } \tau_t^{\text{cand}} < 0.01 \\ \tau_t^{\text{cand}}, & \text{otherwise} \end{cases} \quad (15)$$

When the candidate turnover τ_t^{cand} exceeds the 1% no-trade band ($\tau_t \geq 0.01$), proportional transaction costs $C_t = V_t \cdot c_{\text{bps}} \cdot \tau_t$ are deducted at a flat rate of $c_{\text{bps}} = 0.0015$ (15 bps). Portfolio valuation follows an institutional Open-to-Open execution convention:

$$V_{t+1} = (V_t - C_t)\left[w_{t,0}\left(1 + y_t^{\text{cash}}\right) + \sum_{i=1}^{K} w_{t,i}\frac{O_{t+2,i}}{O_{t+1,i}}\right] \quad (16)$$

where $O_{t+1,i}$ and $O_{t+2,i}$ denote the daily opening prices on the order execution date and the subsequent valuation date, respectively, and $y_t^{\text{cash}} = (1 + \text{FF}_t)^{1/252} - 1$ represents the daily compounded cash yield derived from the effective federal funds rate FF_t.

F. Reward Formulation

The immediate reward function R_t is structured to incentivize net excess return (alpha) over an Equal-Weight (EqW) risky benchmark while explicitly penalizing portfolio turnover:

$$R_t = 100 \times \left(\frac{V_{t+1} - V_t}{V_t} - r_t^{\text{EqW}}\right) - 100 \times \lambda_{\text{turn}} \cdot \tau_t \quad (17)$$

where

$$r_t^{\text{EqW}} = \frac{1}{K}\sum_{i=1}^{K}\left(\frac{O_{t+2,i}}{O_{t+1,i}} - 1.0\right) \quad (18)$$

and the turnover penalty coefficient is set to $\lambda_{\text{turn}} = 0.0015$.

III. Data Pipeline and Temporal Splitting Schemes

The historical dataset spans June 15, 2001, to July 28, 2026, encompassing 6,321 complete trading days. To construct a survivorship-bias-free multi-asset history prior to ETF inception dates, pre-inception price series for GLD are backfilled using continuous COMEX Gold Futures (GC=F) net of a 0.40% annualized expense ratio, while TLT is backfilled using the Vanguard Long-Term Treasury Fund (VUSTX).

A. Feature Engineering

The 17-dimensional per-asset feature vector comprises three domain categories:

1) *Technical Indicators (9):* 14-day RSI, Relative RSI, MACD Histogram Percentage, EMA 12/26 ratio, EMA 50/200 ratio, Price-to-EMA200 ratio, Stochastic RSI %K, 20-day Momentum, and 63-day Momentum.

2) *Cross-Sectional Signals (3):* QQQ/SPY price ratio, SPY/TLT price ratio, and the 63-day Cross-Sectional Momentum Percentile Rank.

3) *Macroeconomic Regime Indicators (5):* Log-transformed CBOE Volatility Index (ln(VIX)), Treasury yield curve slope (TNX − IRX), daily Gold futures return (GC=F), daily WTI Crude Oil futures return (CL=F), and the effective federal funds rate (FEDFUNDS, lagged by 35 calendar days to reflect official publication latency).

Table I lists the partition boundaries of each scheme, and Fig. 1 visualizes the resulting timelines.

TABLE I. Temporal Data-Splitting Timeline and Partition Lengths

Scheme	Fold	Training Period	Validation Period	Test Period (OOS)
Single Holdout	1	2001-06-15 – 2019-12-31 (4,668)	2020-01-02 – 2023-12-29 (1,006)	2024-01-02 – 2026-07-28 (647)
Expanding Window	1	2001-06-15 – 2013-12-31 (3,158)	2014-01-02 – 2015-12-31 (504)	2016-01-04 – 2017-12-29 (503)
	2	2001-06-15 – 2015-12-31 (3,662)	2016-01-04 – 2017-12-29 (503)	2018-01-02 – 2019-12-31 (503)
	3	2001-06-15 – 2017-12-29 (4,165)	2018-01-02 – 2019-12-31 (503)	2020-01-02 – 2021-12-31 (505)
	4	2001-06-15 – 2019-12-31 (4,668)	2020-01-02 – 2021-12-31 (505)	2022-01-03 – 2023-12-29 (501)
	5	2001-06-15 – 2021-12-31 (5,173)	2022-01-03 – 2023-12-29 (501)	2024-01-02 – 2026-07-28 (647)
Sliding Window	1	2001-06-15 – 2013-12-31 (3,158)	2014-01-02 – 2015-12-31 (504)	2016-01-04 – 2017-12-29 (503)
	2	2002-01-02 – 2015-12-31 (3,527)	2016-01-04 – 2017-12-29 (503)	2018-01-02 – 2019-12-31 (503)
	3	2004-01-02 – 2017-12-29 (3,526)	2018-01-02 – 2019-12-31 (503)	2020-01-02 – 2021-12-31 (505)
	4	2006-01-03 – 2019-12-31 (3,525)	2020-01-02 – 2021-12-31 (505)	2022-01-03 – 2023-12-29 (501)
	5	2008-01-02 – 2021-12-31 (3,526)	2022-01-03 – 2023-12-29 (501)	2024-01-02 – 2026-07-28 (647)

Number of trading days in parentheses.

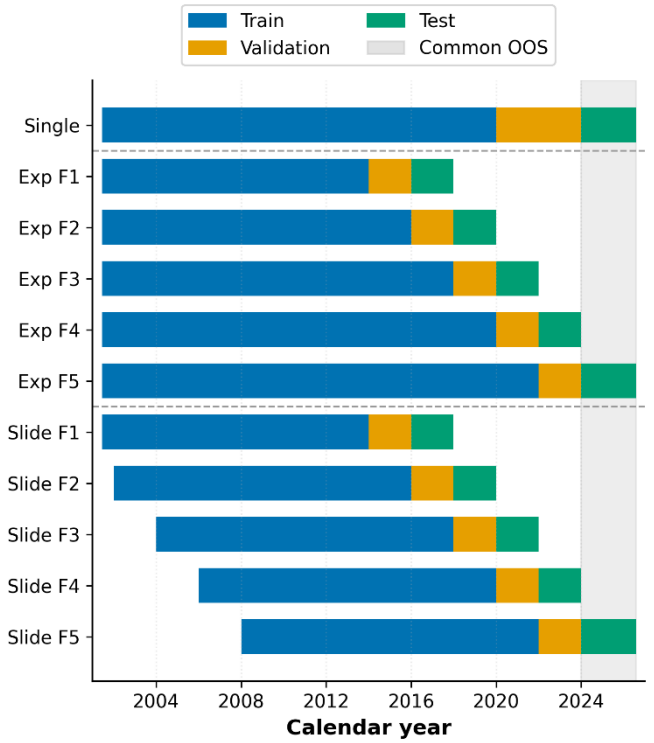


Fig. 1. Temporal data-splitting schemes and the common out-of-sample evaluation horizon. The Common OOS period (2024–2026) represents an identical test partition shared across Single Holdout, Expanding Window Fold 5, and Sliding Window Fold 5.

IV. Dual-Phase System Architecture

To ensure institutional-grade reproducibility and prevent operational contamination between research and production, the system architecture is decoupled into two isolated

subsystems communicating strictly via an immutable artifact interface.

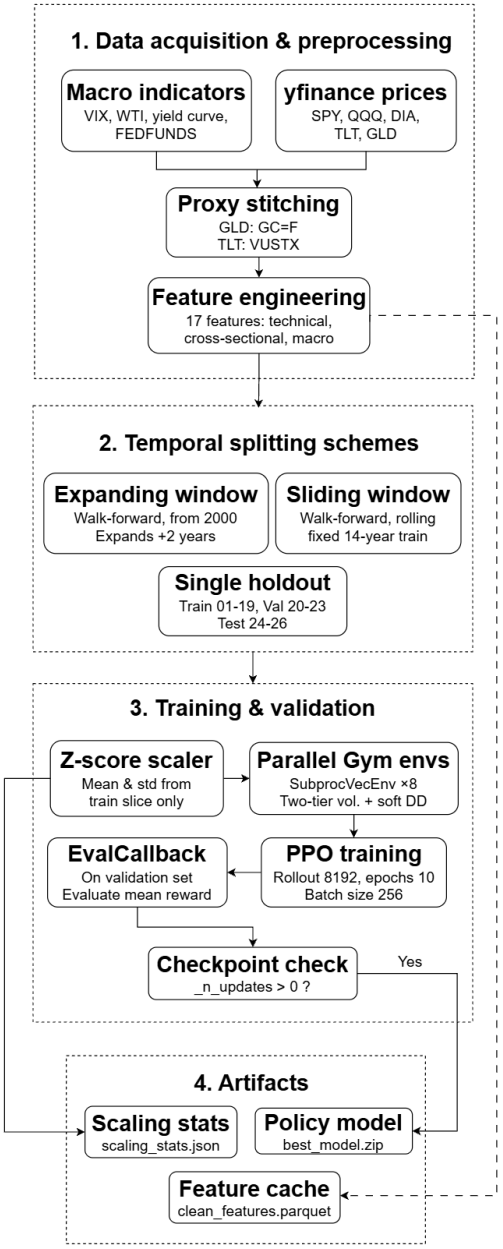


Fig. 2. Offline training and validation pipeline architecture.

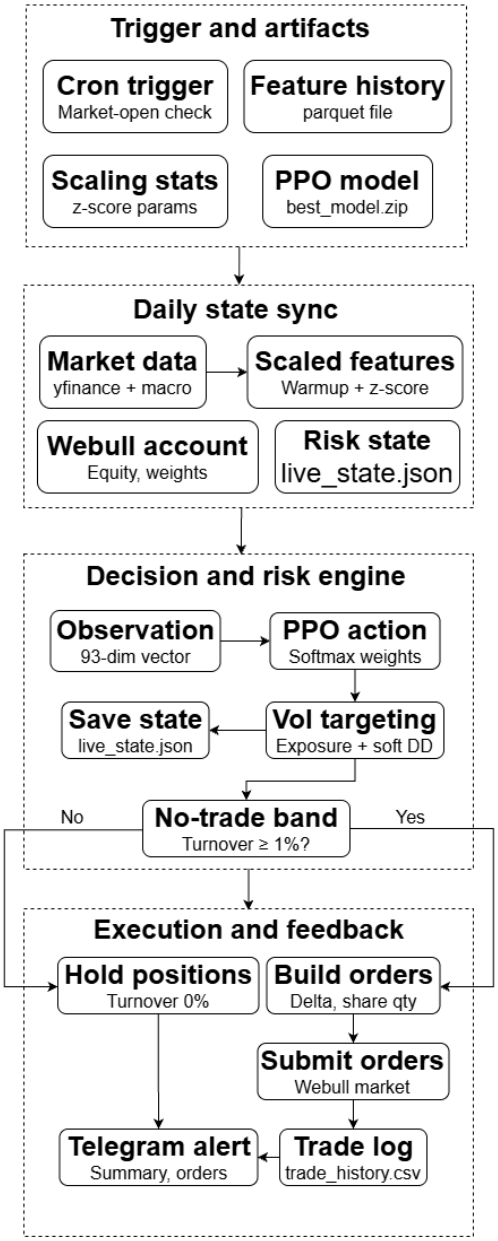


Fig. 3. Online inference, risk overlay, and live execution engine architecture.

The Offline Pipeline (Fig. 2) ingests historical market data, executes proxy stitching, computes Z-score scaling parameters strictly on training slices, and trains the PPO agent across 8 parallel environments (`SubprocVecEnv`) with rollout length 8,192, batch size 256, and 10 optimization epochs. Upon completion, three immutable artifacts are exported to the Online Engine (Fig. 3):

1) `best_model.zip`: The serialized neural network policy weights utilized for deterministic inference.
2) `scaling_stats.json`: The fold-specific training moments $(\mu_j^{\text{train}}, \sigma_j^{\text{train}})$ required to standardize live market observations consistently with the training distribution.
3) `clean_features_full_history.parquet`: The historical feature cache necessary to compute long-horizon warmup indicators (e.g., 200-day moving averages) prior to market open.

During live forward testing, the Online Engine relies on `state_manager.py` to serialize and persist daily risk states within `live_state_<scheme>_ppo.json`, ensuring uninterrupted continuity of the EWMA variance estimate (σ_t^2) and portfolio high-water mark (V_t^{peak}) across trading sessions prior to transmitting market orders to the Webull OpenAPI Sandbox.

V. Empirical Results

A. *Common Out-of-Sample Evaluation (Common OOS: 2024–2026)*

The backtest performance across the shared out-of-sample evaluation window (January 2, 2024 to July 28, 2026; 647 trading days) is summarized in Table II.

TABLE II. Comparative Backtest Performance on the Common Out-of-Sample Period (2024–2026)

Metric	Single Holdout	Expanding (F5)	Sliding (F5)	EqW-VT	EqW-Raw
CAGR (%)	21.93%	22.22%	27.91%	12.24%	16.41%
Sharpe Ratio	1.99	2.01	2.44	1.48	1.34
MDD (%)	−9.77%	−9.77%	−9.11%	−7.03%	−11.57%
Sortino Ratio	1.86	1.87	2.26	—	—
Win Rate (%)	58.45%	58.76%	60.93%	—	—
Profit Factor	1.40	1.41	1.52	—	—
Daily Exp. (%)	+0.0008%	+0.0008%	+0.0010%	—	—
Cash Ratio (%)	7.85%	7.83%	7.89%	—	—

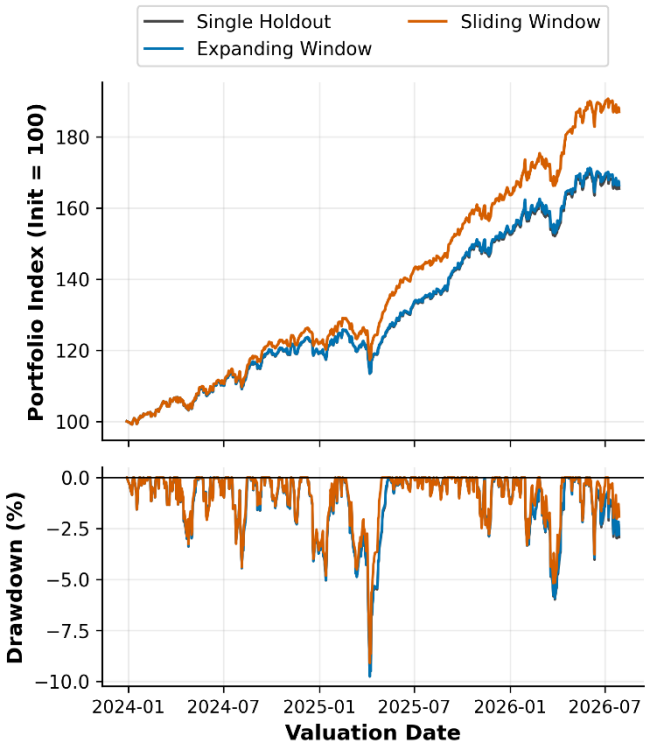


Fig. 4. Cumulative portfolio index and underwater drawdown trajectories during the Common OOS period (2024–2026).

As illustrated in Fig. 4, the Sliding Window scheme purportedly generates substantial excess return, outperforming the Volatility-Targeted Equal-Weight benchmark (EqW-VT) by +15.67% in annualized terms (CAGR of 27.91% vs. 12.24%). Conversely, the Single Holdout and Expanding Window equity curves track each other closely throughout the evaluation horizon, yielding nearly identical Sharpe Ratios of 1.99 and 2.01, respectively.

B. *Cross-Fold Walk-Forward Robustness (2016–2026)*

Table III details the out-of-sample performance across all five walk-forward folds comparing the Expanding Window and Sliding Window architectures.

TABLE III. Cross-Fold Walk-Forward Performance Comparison (Folds 1–5: 2016–2026)

Fold	Test Period	Exp. Sharpe	Slide Sharpe	Exp. MDD (%)	Slide MDD (%)	Dominant Market Regime
1	2016–2017	2.09	2.09	−5.56%	−5.56%	Low-Vol Bull
2	2018–2019	1.01	1.06	−10.80%	−10.59%	Trade War
3	2020–2021	0.38	0.58	−13.76%	−13.67%	COVID-19
4	2022–2023	−0.09	0.18	−31.60%	−26.83%	Rate Hikes
5	2024–2026	2.01	2.44	−9.77%	−9.11%	AI Bull Run
Mean	—	1.08	1.27	−14.30%	−13.15%	—
Stitched	2016–2026	1.01	1.22	−32.31%	−27.52%	—

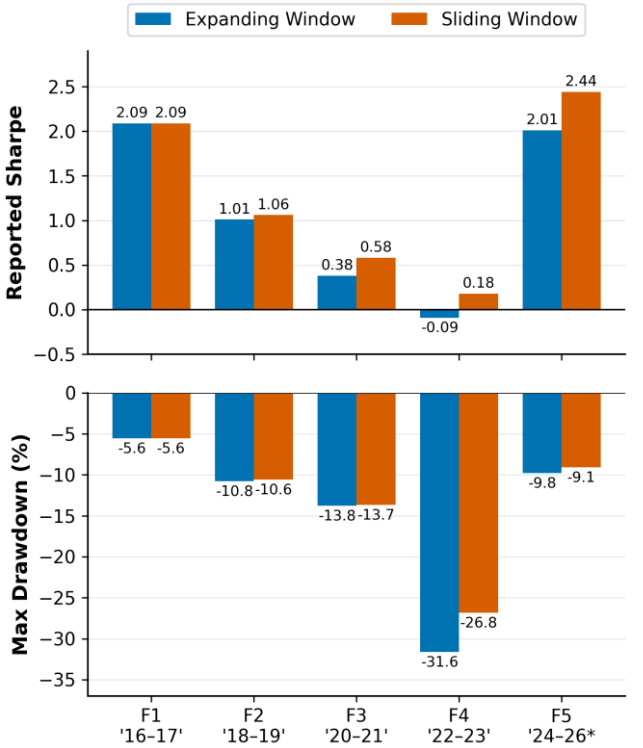


Fig. 5. Out-of-sample Sharpe Ratio and Maximum Drawdown comparison across Folds 1–5.

Fold 4 (2022–2023) represents the most severe stress test for multi-asset portfolios, characterized by a simultaneous sell-off in both equities (SPY, QQQ) and long-duration Treasuries (TLT) triggered by persistent inflation and rapid Federal Reserve monetary tightening. During this regime break, the Sliding Window scheme demonstrates pronounced resilience, maintaining a positive Sharpe Ratio of 0.18 and containing maximum drawdown to −26.83%, whereas the Expanding Window degrades to a negative Sharpe Ratio (−0.09) with a −31.60% drawdown. By systematically discarding obsolete early-2000s observations, the rolling 14-

year window enables faster adaptation to high-interest-rate regimes than an anchored window burdened by multi-decade historical inertia.

VI. Forward Test and Live Execution

A live paper-trading forward test was conducted over five consecutive trading days from October 5 to October 9, 2026, via the Webull OpenAPI Sandbox. The daily account valuations and execution logs are presented in Table IV.

TABLE IV. Daily Live Execution and Account Summary During Forward Testing (October 5–9, 2026)

Valuation Date	Evaluation Metric	Single Holdout PPO	Expanding Window PPO	Sliding Window PPO
2026-10-05	Net Liquidation	$999,992.29	$999,983.04	$1,000,000.00
	Day 1 Order Execution	Buy 5 ETFs (~20% each)	Buy 5 ETFs (~20% each)	Buy 5 ETFs (~20% each)
	Realized Turnover	100.00%	100.00%	100.00%
2026-10-06	Net Liquidation	$1,008,249.25	$1,008,602.50	$1,007,253.94
	Day P/L	+$5,731.62	+$6,092.57	+$4,803.61
	Order Status	No-Trade (0.0% Turn.)	No-Trade (0.0% Turn.)	No-Trade (0.0% Turn.)
2026-10-07	Net Liquidation	$1,000,567.13	$1,000,457.19	$1,000,519.00
	Day P/L	−$6,821.72	−$6,879.32	−$6,784.78
2026-10-08	Net Liquidation	$1,002,132.95	$1,001,929.33	$1,001,961.35
	Day P/L	+$819.20	+$703.55	+$765.83
2026-10-09	Ending Net Liquidation	$1,008,544.74	$1,008,510.25	$1,008,458.66
	Cumulative 5-Day P/L	+$8,552.45	+$8,527.21	+$8,458.66
	Total Return (%)	+0.8553%	+0.8527%	+0.8459%

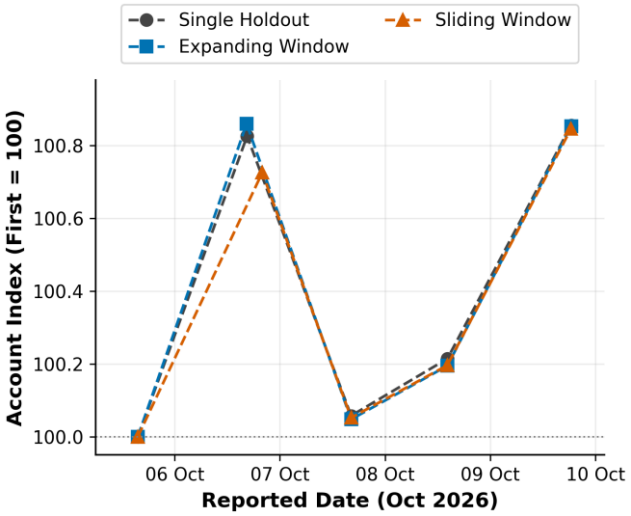


Fig. 6. Normalized account equity index trajectories during the 5-day live forward test (October 2026).

A. Discovery of Policy Degeneracy and Inertia

1) *Immediate Convergence to Equal Weights:* On the initial trading session (October 5), all three deployed models transmitted buy orders allocating almost exactly 20% to each of the five risky ETFs (e.g., Single Holdout allocated 19.9% SPY, 20.1% QQQ, 20.0% DIA, 20.1% TLT, and 19.9% GLD), leaving a negligible residual cash balance of ~0.1%.

2) *Zero Rebalancing Inertia (Days 2–5):* Following the initial deployment, no model executed a single trade across the remainder of the week (Turnover = 0.00%). Day-to-day variations in the raw policy logits were minuscule, failing to breach the `min_turnover_threshold = 0.01` (1% no-trade band). Consequently, all three DRL strategies degenerated into static Equal-Weight Buy-and-Hold portfolios.

VII. Forensic System Audit

A rigorous forensic inspection of the underlying codebase and serialized model checkpoints revealed three critical architectural defects that explain both the anomalous backtest alpha and the live policy degeneracy.

A. The Cash-Yield Normalization Bug

Within the data preprocessing pipeline, the macroeconomic series `fed_funds_rate` was appended to the global `ALL_FEATURES` array and subjected to Z-score standardization via `apply_scaling()`:

$$z_{t,\mathrm{FED}} = \frac{\mathrm{FF}_t - \mu_{\mathrm{FED}}^{\mathrm{train}}}{\sigma_{\mathrm{FED}}^{\mathrm{train}}} \tag{19}$$

However, inside the `InstitutionalPortfolioEnv` class, the `_build_matrices()` method inadvertently extracted this Z-scored feature column rather than the raw interest rate series to compute daily cash returns:

$$y_t^{\mathrm{cash,corrupted}} = \left(1.0 + z_{t,\mathrm{FED}}\right)^{1/252} - 1.0 \tag{20}$$

Statistical Impact: Because $z_{t,\mathrm{FED}}$ assumes negative values whenever the prevailing interest rate falls below its historical training mean and exceeds 1.0 during rate-hiking cycles, the resulting cash yield was severely distorted from economic reality. Crucially, because $\mu_{\mathrm{FED}}^{\mathrm{train}}$ and $\sigma_{\mathrm{FED}}^{\mathrm{train}}$ vary as a function of training window length, identical cash holdings accrued entirely different synthetic interest rates across the three splitting schemes on the exact same calendar date. Since the Sliding Window (Fold 5) training set excluded the higher-rate 2001–2007 regime, its lower training mean inflated $z_{t,\mathrm{FED}}$ during the 2024–2026 high-rate period, artificially boosting its reported Common OOS CAGR and Sharpe Ratio.

B. Premature Checkpoint Selection Trap

Forensic inspection of the internal metadata encapsulated within `best_model.zip` across all three schemes uncovered a striking anomaly:

- `num_timesteps = 4,000`
- `_n_updates = 0`

Technical Root Cause: In the training script, the evaluation callback frequency was parameterized as `eval_freq = max(500, 2000 // max_cpu) = 500` steps per worker. When executed across 8 parallel environments (`SubprocVecEnv`) [5], `EvalCallback` was triggered at a cumulative environment step count of $500 \times 8 = 4{,}000$ timesteps. Because the PPO rollout buffer requires `n_steps = 8,192` per environment (or a full rollout collection) prior to executing its first backpropagation pass, the first evaluation occurred before a single policy gradient update had taken place (`_n_updates = 0`). The callback evaluated the randomly initialized neural network weights,

recorded a baseline validation score, and serialized the untrained network as `best_model.zip`. Subsequent training updates likely degraded validation reward due to noisy reward signals from the corrupted cash yield, preventing later checkpoints from overwriting the initial file. This finding conclusively explains why all live agents output near-zero uniform logits that mapped via Softmax to an exact 20% Equal-Weight allocation.

C. Lookahead Bias in the Volatility-Targeted Baseline

Within `run_ablation_baselines()`, the risk exposure scalar for the EqW-VT benchmark was computed by updating the EWMA variance estimator with the realized return of the current step (`portfolio_step_return = raw_ret`) prior to scaling that exact same day's portfolio return. This contemporaneous coupling introduces single-period lookahead bias into the baseline, rendering EqW-VT an invalid benchmark for strict causal comparison.

VIII. Discussion and Engineering Guidelines

A. Synthesis of Research Questions

- **Response to RQ1:** While the initial Common OOS backtest reported superior performance for the Sliding Window scheme (Sharpe Ratio of 2.44 vs. 2.01 and 1.99), our forensic audit demonstrates that this divergence is an artifact of fold-dependent Z-score distortion in the cash accounting module rather than genuine policy superiority.
- **Response to RQ2:** Walk-forward analysis across Folds 1–5 confirms that rolling fixed-length windows (Sliding Window) provide structural resilience during macroeconomic regime shifts—most notably during the 2022 interest rate shock—by purging obsolete historical regimes from feature scaling and environment dynamics.
- **Response to RQ3:** The 5-day live forward test yielded uniform gains (~0.85%) across all schemes, but empirical inspection of order logs refutes active policy agency. All models suffered from premature checkpointing (`_n_updates = 0`), collapsing into static Equal-Weight portfolios.

B. Engineering Guidelines for Financial DRL Systems

1) *Strict Signal Decoupling:* Raw financial accounting variables (e.g., risk-free interest rates, dividend yields, execution prices, and corporate actions) must be strictly isolated in immutable data structures and never routed through feature normalization pipelines.
2) *Callback Synchronization in Vectorized Environments:* When training on-policy algorithms in multiprocessing environments (`SubprocVecEnv`), `eval_freq` must be parameterized as a strict integer multiple of the rollout horizon (`n_steps`) to guarantee that `EvalCallback` is invoked only after policy gradient updates have occurred.
3) *Automated Checkpoint Auditing:* Deployment pipelines must enforce an automated pre-flight assertion verifying that loaded model artifacts satisfy `_n_updates > 0` and exhibit non-degenerate action variance across synthetic test states prior to live capital allocation.

IX. Conclusion and Data Availability

This study executed an end-to-end quantitative research lifecycle, spanning mathematical MDP formulation, multi-scheme walk-forward backtesting, live broker sandbox execution, and post-mortem forensic auditing. While surface-level backtest metrics suggested that a Sliding Window scheme delivers superior cross-regime adaptability, our forensic audit highlights a foundational axiom of financial machine learning: the rigorous verification of financial accounting integrity and training loop synchronization must strictly precede the interpretation of algorithmic alpha.

A. Reproducibility and Open Artifacts

- *Repository 1 (Research & Offline Training):* https://github.com/AungMoonLord/Trading-Algorithm-Seminar-/tree/main/Work11
- *Repository 2 (Webull Live Execution Engine):* https://github.com/AungMoonLord/Trading-Algorithm-Seminar-/tree/main/Work13
- *Live Inference Infrastructure:* Executed via the Webull OpenAPI Sandbox (api.sandbox.webull.com) integrated with automated real-time Telegram execution reporting.
- *Funding and Conflicts of Interest:* This research received no external funding, and the authors declare no conflicts of interest.

References

[1] X.-Y. Liu, H. Yang, Q. Chen, R. Zhang, L. Yang, B. Xiao, and C. D. Wang, "FinRL: A deep reinforcement learning library for automated stock trading in quantitative finance," 2020, *arXiv:2011.09607*. [Online]. Available: https://arxiv.org/abs/2011.09607

[2] D. H. Bailey and M. López de Prado, "The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting and non-normality," *J. Portfolio Manage.*, vol. 40, no. 5, pp. 94–107, 2014.

[3] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*. [Online]. Available: https://arxiv.org/abs/1707.06347

[4] M. López de Prado, *Advances in Financial Machine Learning*. Hoboken, NJ, USA: Wiley, 2018.

[5] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dormann, "Stable-Baselines3: Reliable reinforcement learning implementations," *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.